
System Overview

The Smart Seed system comprises three integrated components (Figure 1). Component 1 is a spatially explicit probabilistic suitability model operating at 30 m resolution across Nandi County. Component 2 is a ward-level recommendation engine that translates spatial outputs into actionable agronomic guidance. Component 3 is a Web-GIS visualisation and USSD/SMS delivery platform. This section describes the methodology for Components 1 and 2 in full; Component 3 is described in Section 4.3.

Materials and Methods

4.1 Suitability Model and Recommendation Engine

4.1.1 Conceptual Framework

Maize land suitability was evaluated based on Liebig's Law of the Minimum — which holds that crop growth is governed by the single most deficient resource rather than the total available (de Wit, 1992) — combined with fuzzy membership functions calibrated to maize agronomic thresholds following Sys et al. (1993). This framework has been widely adopted in land suitability evaluation studies based on meteorological and soil variables (Zabel et al. 2014; Cronin et al., 2020; Ayambda and Adjei-Gyapong, 2022) and was extended by Zabel et al. (2025) in CropSuite v1.0 to include inter-annual climate variability via a crop failure recurrence rate. Here, we adapt CropSuite for reliable use as the basis of a seed and fertiliser recommender system for Nandi County, Kenya through the following contributions:

(i) Adaptation of the global agronomic thresholds of Sys et al. (1993) - derived from broad international data - to Kenyan conditions using soil nutrient critical levels from KALRO (NAAIAP, 2014).

(ii) Conditioning of crop failure probability on ENSO phase via a Bayesian Beta-Binomial model, drawing on 44 years of historical records (1981–2024). In East Africa, ENSO is among the strongest drivers of inter-seasonal rainfall variability (Nicholson and Kim, 1997), meaning that the long-run failure risk at any given location differs substantially between El Niño, La Niña, and Neutral years; phase-conditioning produces failure probability estimates that better reflect this structure and therefore more reliably inform planting and input decisions.

(iii) Propagation of per-pixel iSDA soil prediction uncertainty through the Liebig minimum via Monte Carlo simulation, producing spatially explicit confidence estimates alongside each suitability classification. Where soil prediction uncertainty is high, a suitability score carries less practical weight; providing uncertainty estimates alongside the classification allows extension officers and farmers to identify locations where recommendations should be treated with greater caution and where targeted soil sampling would most improve guidance quality.

(iv) Translation of probabilistic suitability outputs into ward-level agronomic recommendations covering seed variety selection, expected yield estimation with and without fertiliser, and fertiliser prescription calibrated to Kenyan agronomic data. These are then disseminated in a clear manner via the Web-GIS and USSD/SMS platform described in Sections 4.2 and 4.3.

The suitability model was implemented at 30 m spatial resolution across the full county extent using tiled Monte Carlo simulation ($N = 100$ due to RAM constraints); ward-level outputs were subsequently passed to the recommendation engine described in Section 4.1.8.

4.1.2 Formulation

Under Liebig's principle, overall suitability at pixel p is determined by the single most limiting factor (de Wit, 1992):

$$S^p = \min_{i \in I} \mu_i(x_i^p) \quad (1)$$

where I is the full set of input variables and $\mu_i(\cdot)$ is the piecewise-linear membership function mapping physical values to suitability scores on $[0, 1]$ (Sys et al., 1993).

When x_i^p is a satellite-derived prediction rather than a direct measurement, uncertainty in the true value propagates through the minimum operator. It follows that $E[\min_i \mu_i] \neq \min_i E[\mu_i]$ by Jensen's inequality: the expected minimum is systematically lower than the minimum of expected values, meaning deterministic use of predicted soil means is expected to produce an optimistic bias. Monte Carlo estimation of $E[\min_i \mu_i]$ addresses this bias:

$$\bar{S}^p = (1/N) \sum_{n=1}^N [\min_{i \in I} \mu_i(x_i^{p,(n)})] \quad (2)$$

where $x_i^{p,(n)}$ is the n -th sample from the uncertainty distribution for variable i at pixel p . All variables are sampled independently across N Monte Carlo iterations; known inter-variable correlations — such as between pH, calcium, and base saturation, or between clay content and CEC — are not modelled, which represents a structural simplification whose directional effect on the Liebig minimum is location-dependent and indeterminate.

4.1.2 Spatial Reference Framework

All raster data were co-registered to a common 30 m spatial reference grid derived from the iSDA soil pH (0–20 cm) layer (Hengl et al., 2021). The affine transform, coordinate reference system, and raster dimensions from this reference file were applied uniformly to all outputs. A county-level validity mask was constructed from the Nandi County administrative boundary (OCHA, 2020) using `rasterio.features.geometry_mask` with `all_touched = True`, ensuring complete coverage of boundary pixels. All pixels falling outside the county boundary were assigned NaN values and excluded from all calculations. Ward boundaries, used for spatial aggregation of model outputs, were obtained from the same source.

4.1.3 Input Data

Soil Data

Soil property data were sourced from the iSDA Africa 30 m satellite-derived soil maps (Hengl et al., 2021), which provide spatially continuous mean and standard deviation rasters for each property derived from ensemble machine learning. The standard deviation rasters represent prediction uncertainty and were used to parameterise stochastic sampling within the Monte Carlo framework. Fourteen soil properties were extracted: soil pH, organic carbon, total nitrogen, extractable calcium, magnesium, potassium, and phosphorus, extractable zinc and iron, effective cation exchange capacity (ECEC), clay content, stone content, depth to bedrock, and soil texture class (Table A1). Texture class was treated as a categorical variable and converted to suitability scores via a fixed lookup table (Table A2); all remaining properties were treated as continuous variables subject to stochastic sampling.

Prediction reliability stratification. Variables were stratified into three reliability tiers based on published Concordance Correlation Coefficients (CCC; Lin, 1989) from iSDA's continental cross-validation (Hengl et al., 2021; Table 1). High-reliability variables ($CCC \geq 0.85$) enter the Liebig minimum with their native uncertainty distributions; medium-reliability variables ($0.75 \leq CCC < 0.85$) are included but outputs are flagged for moderate uncertainty. Advisory variables ($CCC < 0.75$) — extractable phosphorus, stone content, and depth to bedrock — enter the Liebig minimum using published iSDA prediction intervals but are also reported as separate advisory rasters, as published intervals may underestimate true uncertainty for low-CCC variables; exclusion was not adopted as these are potentially limiting factors in Nandi County soils (NAAIAP, 2014).

Salinity and sodicity variables were considered but excluded. All soils in Nandi fall within the non-saline class ($E_{ce} < 2 \text{ dS m}^{-1}$) of the Global Soil Salinity Map (Ivushkin et al., 2019), meaning salinity is never a limiting factor and its omission introduces no bias; exchangeable sodium data from Africa SoilGrids (Hengl et al., 2015) were additionally found to carry insufficient spatial precision for pixel-level suitability scoring, with a continental RMSE of $3.61 \text{ cmol}^+ \text{ kg}^{-1}$ against a Nandi county range of approximately $6 \text{ cmol}^+ \text{ kg}^{-1}$.

Table 1. Concordance Correlation Coefficients (CCC) for iSDA soil variables (Hengl et al., 2021) and reliability tier assignment. Colour coding: green = High; grey = Medium; orange = Advisory.

Variable	CCC	Tier
pH in H ₂ O	0.900	High
Organic carbon	0.883	High
Effective CEC	0.860	High
Calcium, extractable	0.913	High
Magnesium, extractable	0.898	High
Potassium, extractable	0.872	High
Iron, extractable	0.899	High
Clay content	0.854	Medium
Total nitrogen	0.845	Medium

Zinc, extractable	0.831	Medium
Extractable phosphorus	0.654	Advisory
Stone content	0.701	Advisory
Depth to bedrock	0.725	Advisory

Decoding. iSDA soil means are stored as log-encoded integers and were decoded as $\mu = \text{expm1}(\text{raw}/s)$, where $s = 10$ for most variables and $s = 100$ for total nitrogen, which uses finer precision encoding (Hengl et al., 2021). Soil pH and depth to bedrock use linear encoding and were decoded by multiplying by 0.1 and 1.0 respectively. Standard deviation rasters are stored on a linear scale and were never passed through the inverse log transform. Following decoding, extractable calcium, magnesium, and potassium (mg kg^{-1}) were converted to cmol kg^{-1} by dividing by molar masses of 200.39, 121.53, and $390.98 \text{ g mol}^{-1}$ respectively, accounting for valency. Organic carbon and total nitrogen were converted from g kg^{-1} to percentage by multiplying by 0.1.

Depth treatment. Sys et al. (1993) specify depth-dependent weighting rules for soil properties based on their agronomic relevance at different horizons; these were followed as described below. ECEC was taken from the 20–50 cm (B-horizon) layer only, as recommended for uniform soil profiles where the subsoil better represents permanent structural cation exchange capacity independent of organic matter inputs. Stone content and clay content were available at both 0–20 cm and 20–50 cm; Sys et al. (1993) recommend standard depth weighting where stone content is homogeneous or increases with depth, and analysis of the two depth intervals confirmed this to be the case in Nandi County — stone content was approximately homogeneous between depths and clay content increased with depth while silt content remained approximately constant (Fig. A1). The depth-weighting scheme of Sys et al. (1991) was therefore applied:

$$\mu_w = (2.0 \times \mu_{\text{top}} + 1.5 \times \mu_{\text{sub}}) / 3.5 \quad (3)$$

Variance in the weighted mean was propagated as:

$$\sigma_w = \sqrt{(4.0\sigma_{\text{top}}^2 + 2.25\sigma_{\text{sub}}^2) / 3.5} \quad (4)$$

All remaining soil variables were sourced from the 0–20 cm layer, consistent with the Sys et al. (1993) recommendation to use the upper horizon for pH, cations, and organic carbon.

Derived variable: Apparent CEC. Apparent cation exchange capacity ($\text{CEC}_{\text{apparent}}$) was computed within the Monte Carlo loop following Sys et al. (1993):

$$\text{CEC}_{\text{apparent}} = ([\text{ECEC} - (2.6 \times \%OC)] / \% \text{clay}) \times 100 \quad (5)$$

where ECEC is in cmol kg^{-1} and OC is organic carbon in percentage. Soil texture class was converted to suitability scores via a lookup table mapping each of the 12 USDA texture classes to a fixed score derived from Sys et al. (1993) (Table A2), with depth-weighted scores computed using the 2.0/1.5 scheme.

Terrain Data

Elevation data were sourced from the NASA SRTM 30 m digital elevation model (NASA JPL, 2013) and used both the reference surface for lapse-rate temperature correction and as an input to seed variety elevation filtering. A pre-derived slope raster (percentage gradient) was loaded at native 30 m resolution. Given the relatively coarse breakpoints of the slope membership function employed — suitability changes only at 4%, 8%, 16%, 30%, and 50% — slope was treated as deterministic, as it is so small DEM-derived errors are unlikely to move a pixel across any of these thresholds.

Climate Data

Climate data were extracted from Google Earth Engine using the ECMWF ERA5-Land daily aggregate product (~9 km resolution; Muñoz-Sabater et al., 2021) for temperature and relative humidity, and CHIRPS daily precipitation (~5 km resolution; Funk et al., 2015). The Long Rains season was defined as the 110-day window from 25 March to 13 July based on reported optimal planting times in Nandi (CropGenius Africa, 2025) and an average maize growing cycle of 110 days (Zabel et al., 2025). The Short Rains season was defined from 25 October to 12 February of the following calendar year. Ten-year mean seasonal values (2015–2024) were used to characterise recent climatological conditions across both seasons.

The following variables were extracted per season. Mean seasonal rainfall (mm) and monthly rainfall totals for four successive 30-day blocks were obtained from CHIRPS over the full 110-day window. Mean, minimum, and maximum daily temperatures (°C) were obtained from ERA5-Land over the full 110-day window, along with mean germination temperature (°C) computed over the first 30 days following season onset. Relative humidity was extracted at two physiological stages: vegetative development (days 31–60) and grain maturation (days 81–110). ERA5-Land temperatures were converted from Kelvin to Celsius during GEE extraction. Relative humidity was not directly available from ERA5-Land and was computed from 2 m air temperature (T) and dewpoint temperature (T_d) using the August–Roche–Magnus approximation (Alduchov & Eskridge, 1996):

$$RH = \exp(17.625T_d / (T_d + 243.04)) / \exp(17.625T / (T + 243.04)) \times 100 \quad (6)$$

All climate rasters were upsampled to 30 m using bilinear interpolation (order=1). A lapse-rate correction is applied to the four temperature variables to account for the strong altitudinal gradient within Nandi County. A reference elevation surface consistent with the ERA5-Land model orography was constructed by: (i) downscaling the 30 m SRTM DEM to the ERA5 grid resolution; (ii) filling any NaN cells in the coarse elevation array with the spatial mean; and (iii) upscaling the filled coarse elevation back to 30 m, producing a smoothed reference surface (e_{ref}) that is spatially self-consistent with the ERA5-Land orography. Each temperature variable was then corrected as:

$$T_{corr} = T_{upscaled} + (z_{DEM} - e_{ref}) \times (-0.0065) \quad (7)$$

where z_{DEM} is the 30 m SRTM elevation (m) and $-0.0065 \text{ } ^\circ\text{C m}^{-1}$ is the standard environmental lapse rate (Barry & Chorley, 2003). It should be noted that climate data were available at approximately 5–9 km resolution and bilinear downscaling to 30 m does not physically resolve sub-pixel meteorological gradients. Climate data were available at approximately 5–9 km resolution; bilinear downscaling to 30 m produces a spatially smooth surface but does not introduce new meteorological information below the native grid scale, and climate-driven suitability differences between neighbouring pixels should be interpreted with this in mind.

Crop Failure Frequency and ENSO Phase Conditioning

Annual crop failure rasters were computed for all 44 growing seasons from 1981 to 2024 using Google Earth Engine. For each year and pixel, a binary failure indicator was assigned if any of the following threshold violations occurred during the growing season window, following the maize-specific lethal threshold parameterisation of Zabel et al. (2025) as implemented in the CropSuite v1.0 model repository (Knüttel & Zabel, 2025):

- Cold stress: minimum daily temperature $< 5^\circ\text{C}$ for ≥ 7 consecutive days
- Heat stress: maximum daily temperature $> 32^\circ\text{C}$ for ≥ 5 consecutive days
- Drought: daily precipitation $< 1 \text{ mm}$ for ≥ 21 consecutive days
- Flooding: cumulative precipitation $> 100 \text{ mm}$ within any 3-day window

Annual five-band GeoTIFF rasters encoding binary failure status for each stressor (cold, heat, drought, flood, and overall failure — defined as any of the four individual stressors being triggered) were exported at approximately ~9 km resolution and upsampled to 30 m using nearest-neighbour interpolation to preserve binary values.

Failure probabilities were conditioned on ENSO phase rather than pooled unconditionally across all years, as we consider that unconditional pooling averages over climatologically distinct phases and may produce estimates that do not reflect the long-run risk experienced during any particular phase. This distinction is particularly relevant in East Africa, where seasonal rainfall is strongly influenced by ENSO (Nicholson & Kim, 1997). ENSO phase classification used the NOAA Oceanic Niño Index (ONI), obtained from the NOAA Climate Prediction Center (NOAA CPC, 2024), with the March–May seasonal average used for Long Rains classification and October–December for Short Rains. Years with $ONI \geq +0.5$ were classified as El Niño (EN), $ONI \leq -0.5$ as La Niña (LN), and remaining years as Neutral (N). Long-run phase probabilities (π^{EN} , π^{LN} , π^N) were estimated from the 44-year record. For each phase s , failure counts k_s^p were accumulated for each of the five risk bands and each pixel across all n_s years of that phase. Under independent

Beta (1,1) priors, the Bayesian posterior for the failure rate in each phase is:

$$p_s^p \sim \text{Beta}(1 + k_s^p, 1 + n_s - k_s^p), s \in \{EN, LN, N\} \quad (8)$$

This conjugate model updates the non-informative uniform prior with the observed binary failure history; a Beta(1,1) fallback was applied for any phase with zero observed years. We reason that entering failure rates directly into the Liebig minimum would conflate two conceptually distinct dimensions of agricultural risk — chronic unsuitability due to unfavourable mean conditions, and acute inter-annual volatility — obscuring the distinction between a ward that is structurally unsuitable and one that is suitable on average but climatically unreliable. The failure bands are therefore reported as independent risk indicators and used to generate ward-level climate risk warnings, as described in Section 4.1.8.

4.1.4 Fuzzy Membership Functions

Suitability thresholds were predominantly derived from Sys et al. (1993) following standard practice (Wanyama et al., 2019; Awoonor et al., 2023; Wang, 2015). Nutrient thresholds for total nitrogen, phosphorus, and exchangeable cations were sourced from Awoonor et al. (2023), cross-referenced against Linda & Oluwatola (2015) for calcium, potassium, and magnesium. Effective CEC suitability values were derived from the critical levels presented in Adamu et al. (2015). Breakpoints for pH, organic carbon, phosphorus, calcium, potassium, and magnesium were further refined using Kenya-specific soil nutrient critical levels from NAAIAP (2017); adjusted values are indicated by an asterisk in Table A1. Maximum temperature thresholds were cross-referenced against Awoonor et al. (2023) and Zabel et al. (2025). Zinc and iron suitability thresholds were derived from critical levels reported by NAAIAP (2017); given the limited empirical basis for a full membership function, a conservative two-point function was applied in which pixels below the critical level receive a score of 0.95 and those above receive 1.0.

Scoring was implemented using `numpy.interp`, which performs linear interpolation within the defined range and clamps to boundary values outside it. Complete membership function breakpoints for all 25 input variables are provided in Table A1.

Three composite climate scores were computed as minimum-of-components before the Monte Carlo loop, as they are deterministic functions of fixed climate rasters: rainfall suitability (minimum of mean seasonal score and four monthly scores); temperature suitability (minimum of mean, minimum, maximum, and germination temperature scores); and relative humidity suitability (minimum of development and maturation stage scores).

4.1.5 Monte Carlo Simulation and Uncertainty Propagation

Soil sampling. For each soil variable, $N = 100$ realisations per pixel were drawn from a distribution parameterised by the iSDA predicted mean and standard deviation. Variables with observed right-skewed distributions (Fig. A2) were modelled as log-normal:

organic carbon, extractable phosphorus, potassium, total nitrogen, calcium, and magnesium. Samples were drawn as:

$$x_{\text{samp}} = \exp[\ln(\mu) + \text{clip}(\sigma / \max(|\mu|, 10^{-8}), 10^{-6}, 5) \cdot Z], \\ Z \sim N(0, 1) \quad (9)$$

All remaining variables (pH, ECEC, stone content, bedrock depth, clay content, zinc, iron) were modelled as normally distributed:

$$x_{\text{samp}} = \mu + \sigma^* \cdot Z \quad (10)$$

where $\sigma^* = 10^{-4}$ if σ is non-finite or below 10^{-4} , and σ otherwise. All samples were clipped to variable-specific physical bounds (e.g. pH: [3.5, 10.0]; organic carbon: [0.0, 20.0%]; extractable phosphorus: [0.0, 200.0 mg kg⁻¹]). All $N_{\text{MC}} \times H_{\text{tile}} \times W_{\text{tile}}$ draws were generated in a single vectorised call to `numpy.random.default_rng.standard_normal`.

ENSO sampling. In each Monte Carlo iteration, an ENSO phase $s^{(n)}$ was sampled from $\text{Categorical}(\pi^{\text{EN}}, \pi^{\text{LN}}, \pi^{\text{N}})$. A phase-conditional failure rate was then drawn for each of the five risk bands and each pixel from the Beta posterior of Equation (8). This, as noted, marginalises over both ENSO phase uncertainty and within-phase binomial sampling uncertainty, rather than pooling failure frequency across the distinct phases. The five failure band rates are retained to generate ward-level climate risk warnings as described in Section 4.1.8, but do not enter the Liebig minimum for the reasons noted in Section 4.1.3.

Computational implementation. The full county raster was processed in non-overlapping 256×256 pixel tiles; all full-resolution arrays were held in RAM throughout, with tile results written back to global output arrays after each tile. All deterministic scores — slope, texture, and composite climate scores — were computed once per tile before the MC loop and reused across all iterations. All soil MC sample arrays (shape: $N_{\text{MC}} \times H_{\text{tile}} \times W_{\text{tile}}$) were scored through their membership functions in a single vectorised `numpy.interp` call before the loop. Apparent CEC was computed from the full stochastic ECEC, OC, and clay arrays simultaneously before the loop.

4.1.6 Suitability Aggregation and Statistics

The suitability for iteration n at pixel p was computed as the Liebig minimum across all 18 scored inputs:

$$S^{p,(n)} = \min(\mu^{p,(n)}_{\text{climate}}, \min_{i \in \text{soil}} \mu^{p,(n)}_i, \mu^{p,(n)}_{\text{slope}}, \mu^{p,(n)}_{\text{texture}}) \quad (11)$$

comprising slope, depth-weighted texture, rainfall composite, temperature composite, relative humidity composite, ECEC, CEC_app, organic carbon, zinc, iron, pH, total nitrogen, calcium, magnesium, potassium, phosphorus, bedrock depth, and stone content. The variable returning the minimum score in each iteration was recorded; the limiting factor frequency was computed as the fraction of iterations in which each variable was the argmin. The posterior mean and standard deviation of suitability were:

$$\bar{S}^p = (1/N) \sum S^{p,(n)} \quad (12)$$

$$\sigma_{S^p} = \sqrt{\max[(1/N) \sum (S^{p,(n)})^2 - (\bar{S}^p)^2, 0]} \quad (13)$$

Per-variable mean scores and standard deviations were computed equivalently across all 100 iterations.

4.1.7 FAO Suitability Classification and Confidence Tiers

The posterior mean $\bar{S}^p$ was classified into four FAO land suitability classes (FAO, 1976; Sys et al., 1993):

S1 (Highly Suitable): $\bar{S}^p > 0.75$. Minimal limitations for maize production.

S2 (Moderately Suitable): $0.50 < \bar{S}^p \leq 0.75$. Moderate limitations requiring management intervention.

S3 (Marginally Suitable): $0.25 < \bar{S}^p \leq 0.50$. Severe limitations; marginal viability under smallholder conditions.

N (Not Suitable): $\bar{S}^p \leq 0.25$. Prohibitive limitations.

Class 0 (No Data): $\bar{S}^p = \text{NaN}$. Outside county boundary or missing input data.

As such, for each season we obtained six primary GeoTIFFs — mean suitability, suitability standard deviation, FAO classification, confidence, a five-band risk probability raster, and a five-band risk uncertainty raster — alongside three advisory rasters for low-reliability soil variables, and per-variable score and raw-value rasters for all input factors. The mean suitability, FAO suitability scores, suitability-associated uncertainty, and limiting factors for Nandi are displayed in Figures A3 and A4 for *Long Rains* and *Short Rains*, respectively.

4.1.8 Ward-Level Recommendation Engine

The above approach supports both ward-level and higher-resolution coordinates-based recommendation. The recommendation in this pilot is applied at the ward-level. As such, we computed the spatial means of all raster outputs over each ward polygon (Figure A5) and stored as attribute columns in a ward-level GeoDataFrame (OCHA, 2020). The recommendation engine then operated entirely on these aggregated values.

Seed variety scoring. Seed variety data (Table A4) - containing optimal precipitation ranges, elevation, key characteristics, and potential yield - was scraped from KenyaSeed. We adopt a weighted scoring system using importances derived by Wanyama et al. (2019) from expert knowledge. For each ward and season, varieties were first filtered to those whose stated elevation range encompassed the ward mean elevation. Each eligible variety was then scored on up to six weighted criteria: precipitation match (weight = 0.47, always active); elevation centrality (0.13, always active); flood resistance (0.47, triggered when ward flood probability exceeded the county mean); drought/moisture-stress tolerance (0.13, triggered when ward drought probability exceeded the county mean); heat tolerance (0.13, triggered when ward maximum temperature exceeded 35°C); and cold tolerance (0.13, triggered

when ward minimum temperature fell below 10°C). Elevation centrality was computed as $1 - |z_{\text{ward}} - z_{\text{mid}}| / z_{\text{half}}$, where z_{mid} is the midpoint and z_{half} the half-range of the variety's elevation range. The final weighted score was earned weight divided by applicable weight. The top three varieties by score were returned; where scores

Yield estimation. Expected yield without fertiliser was estimated by scaling the variety's stated potential yield by fractions determined from ward suitability score: 85–100% for $\bar{S} \geq 85\%$; 60–85% for $60\% \leq \bar{S} < 85\%$; 40–60% for $40\% \leq \bar{S} < 60\%$; and 0–40% for $\bar{S} < 40\%$ (Wanyama et al., 2019; Wang, 2015). Yield with fertiliser was estimated by adding nitrogen and phosphorus response increments using the Mitscherlich response function:

$$\Delta Y = b \times (1 - c^i)$$

with empirically derived OFRA Rift Valley Upper regional coefficients (Kibunja et al., 2017). Separate coefficient sets were used for high-potential varieties ($\geq 3.0 \text{ t ha}^{-1}$: $b_N = 2.640$, $c_N = 0.903$, $b_P = 0.738$, $c_P = 0.904$) and low-potential varieties ($b_N = 0.200$, $c_N = 0.886$, $b_P = 0.792$, $c_P = 0.990$). Total yield increment was capped at the variety's stated potential yield.

Fertiliser recommendations. Fertiliser prescriptions accounted for Kenya-specific critical value¹ data (NAAIAP, 2014) to determine nutrient deficiencies, Rift Valley Upper economically optimal rate (EOR) values (Kibunja et al., 2017) to elucidate recommended fertiliser application rates, and KALRO's *Maize Production and Training Manual* to determine the recommended fertiliser types available in the region (Karanka et al., 2016). From this, we develop a decision tree based (Table A5) on three ward-level soil parameters: pH (critical value 5.5), total nitrogen as percentage (critical value 0.2%), and extractable phosphorus in mg kg^{-1} (critical value 15 mg kg^{-1}). The eight possible combinations produced product and application rate recommendations detailed in Table 5. Two unconditional micro-nutrient checks were appended: zinc sulfate was recommended if the ward mean zinc suitability score was below 1.0; potassium chloride (50 kg ha^{-1} band application) was recommended if ward mean potassium fell below 0.256 cmol kg^{-1} (Kibunja et al., 2017).

Climate risk warnings and output dissemination. If the ward's phase-weighted mean drought probability (Beta-Binomial Band 2) exceeded the county average, a drought risk warning was issued and staged fertiliser application recommended. If ward flood probability (Band 3) exceeded the county average, farmers were warned against basal fertiliser immediately before forecast heavy rainfall.

An example output for *Long Rains* in the Kapkangani ward is displayed in Fig A6.

¹ The *critical value* is defined as the value above which application of fertiliser is not expected to improve yield (NAAIAP, 2014).

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Appendix A: Membership Function Parameters

Table A1. Fuzzy membership function breakpoints for all input variables. Values marked * denote Kenya-specific critical levels from NAAIAP (2017). Variables marked † are advisory-tier (CCC < 0.75) and reported as separate advisory rasters in addition to entering the Liebig minimum. Reference codes are defined in the legend below the table.

Parameter	Source	Values	Suitability Scores	Ref.
Climate and Growing Conditions				
Mean Season Rainfall (mm)	CHIRPS	0, 300, 400, 500, 600, 750, 900, 1200, 1600	0.25, 0.6, 0.85, 0.95, 1.0, 1.0, 0.95, 0.85, 0.60	[1]
Precipitation Month 1 (mm)	CHIRPS	0, 60, 75, 100, 125, 175, 220, 295, 400, 475	0.25, 0.6, 0.85, 0.95, 1.0, 1.0, 0.95, 0.85, 0.6, 0.25	[1]
Precipitation Months 2–3 (mm)	CHIRPS	0, 70, 120, 150, 175, 200, 235, 310, 400, 475	0.25, 0.6, 0.85, 0.95, 1.0, 1.0, 0.95, 0.85, 0.6, 0.25	[1]
Precipitation Month 4 (mm)	CHIRPS	0, 60, 80, 100, 125, 165, 210, 285, 400, 475	0.25, 0.6, 0.85, 0.95, 1.0, 1.0, 0.95, 0.85, 0.6, 0.25	[1]
Mean Temperature (°C)	ERA5-Land	0, 14, 16, 18, 22, 24, 26, 32, 35, 40	0.25, 0.6, 0.85, 0.95, 1.0, 1.0, 0.95, 0.85, 0.6, 0.25	[1]
Germination Temp (°C)	ERA5-Land	0, 10, 13, 18, 21, 24, 29	0.0, 0.4, 0.85, 1.0, 0.85, 0.4, 0.0	[1]
Min Temperature (°C)	ERA5-Land	0, 7, 9, 12, 16, 17, 18, 24, 28, 30	0.25, 0.6, 0.85, 0.95, 1.0, 1.0, 0.95, 0.85, 0.6, 0.25	[1]

Max Temperature (°C)	ERA5-Land	24, 26, 32, 35, 40	1.0, 0.95, 0.85, 0.60, 0.25	[2,3]
RH Development (%)	ERA5-Land	0, 30, 36, 42, 50, 65, 80	0.25, 0.6, 0.85, 0.95, 1.0, 1.0, 0.95	[1]
RH Maturation (%)	ERA5-Land	0, 20, 24, 30, 40, 50, 75, 90	0.6, 0.85, 0.95, 1.0, 1.0, 0.95, 0.85, 0.6	[1]
Crop Failure Frequency	ERA5-Land / CHIRPS	0.0, 0.025, 0.05, 0.075, 0.10, 0.125, 0.15, 0.175, 0.20, 0.225, 0.25	1.0, 0.98, 0.95, 0.88, 0.73, 0.50, 0.27, 0.12, 0.05, 0.02, 0.0	[1,3]
Terrain				
Slope (%)	SRTM	0, 4, 8, 16, 30, 50	1.0, 0.95, 0.85, 0.60, 0.40, 0.25	[1]
Soil Physical Properties				
Soil Texture Class	iSDA	Classes 1–12 (USDA) — see Table A2	—	[1]
Stone Content (vol%) †	iSDA (depth-wtd)	0, 3, 15, 35, 55	1.0, 0.95, 0.85, 0.60, 0.25	[1,2]
Depth to Bedrock (cm) †	iSDA	0, 20, 50, 75, 100	0.25, 0.6, 0.85, 0.95, 1.0	[1]
Soil Chemical Properties				
pH (H ₂ O)	iSDA 0–20cm	0, 5.2, 5.5*, 5.8, 6.2, 6.6, 7.0, 7.8, 8.2, 8.5	0.4, 0.6, 0.85, 0.95, 1.0, 1.0, 0.95, 0.85, 0.6, 0.25	[1]*[4]
Effective CEC (cmol/kg soil)	iSDA 20–50cm	0, 5, 10	0.85, 0.95, 1.0	[5]
Apparent CEC (cmol/kg clay)	iSDA (derived)	0, 10, 16, 24	0.4, 0.85, 0.95, 1.0	[1,2]
Organic Carbon (%) †	iSDA 0–20cm	0, 0.5, 0.7, 1.5, 2.7*	0.4, 0.6, 0.85, 0.95, 1.0	[1]*[4]
Total Nitrogen (%)	iSDA 0–20cm	0, 0.1, 0.15, 0.2	0.6, 0.85, 0.95, 1.0	[2]
Extractable Phosphorus (mg/kg) †	iSDA 0–20cm	0, 6, 30*	0.4, 0.6, 1.0	[2]*[4]
Exchangeable Ca (cmol/kg)	iSDA 0–20cm	0, 1, 2*	0.6, 0.85, 1.0	[2,6]*[4]
Exchangeable K (cmol/kg)	iSDA 0–20cm	0, 0.1, 0.24*	0.85, 0.95, 1.0	[2,6]*[4]
Exchangeable Mg (cmol/kg)	iSDA 0–20cm	0, 0.2, 0.5, 1.0*	0.25, 0.4, 0.6, 1.0	[2,6]*[4]
Zinc (mg/kg)	iSDA 0–20cm	0, 5.0*	0.95, 1.0	[4]
Iron (mg/kg)	iSDA 0–20cm	0, 10.0*	0.95, 1.0	[4]

References:

- [1] Sys et al. (1993) [2] Awoonor et al. (2023) [3] Zabel et al. (2025)
 [4] NAAIAP (2017) [5] Adamu et al. (2015) [6] Linda & Oluwatola (2015)
 * Kenya-specific critical level from NAAIAP (2017)
 † Advisory-tier variable (CCC < 0.75): participates in Liebig minimum but also reported as advisory raster

Depth-weighted variables (stone content, clay content): $\mu_w = (2.0 \times \mu_{top} + 1.5 \times \mu_{bot}) / 3.5$
ECEC sourced from 20–50 cm (B-horizon); all other soil variables from 0–20 cm layer

Table A2. Soil texture class lookup table mapping iSDA integer codes to suitability scores (Sys et al., 1993).

Description	iSDA code	Sys et al. symbol	Suitability
Clay	1	Cm; C±60; Co	0.83
Silty Clay	2	SiCm, SiCs	0.25
Sandy Clay	3	SC	0.85
Clay Loam	4	CL	1.0
Silty Clay Loam	5	SiCL	1.0
Sandy Clay Loam	6	SCL	0.95
Loam	7	L	0.95
Silt Loam	8	SiL	1.0
Sandy Loam	9	SL	0.85
Silt	10	Si	1.0
Loamy Sand	11	LfS; LS	0.77
Sand	12	fS; S	0.6

Add: Qualitative land suitability evaluation for maize (*Zea mays* L.) production on four benchmark soils in the forest-savanna transition agro-ecological zone of Ghana

Fig. A1. Stone content and texture heterogeneity over 0-20cm and 20-50cm depths.

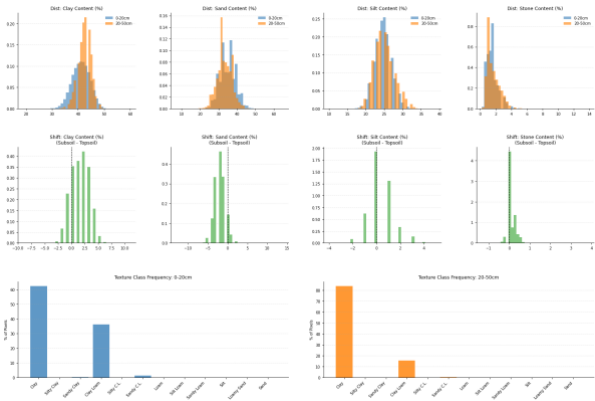


Fig A2. Distribution of soil physical and chemical properties in Nandi County. Soil Texture Class colours represent suitability based on a gradient from red (low suitability) to green (high suitability).

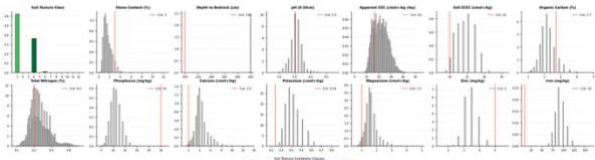


Fig A3. Suitability maps for *Long Rains* season.

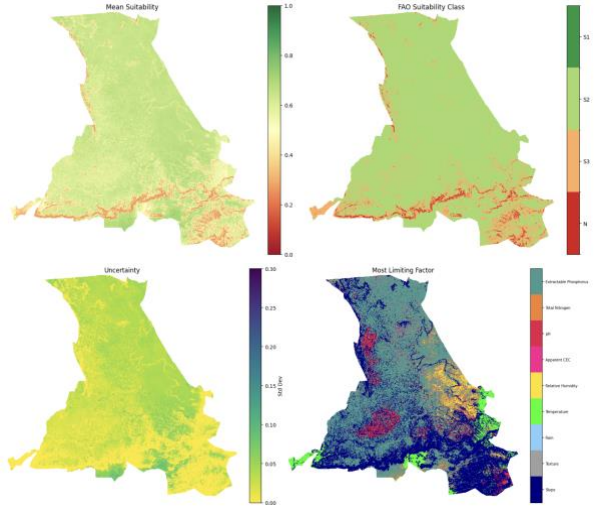


Fig A4. Suitability maps for *Short Rains* season.

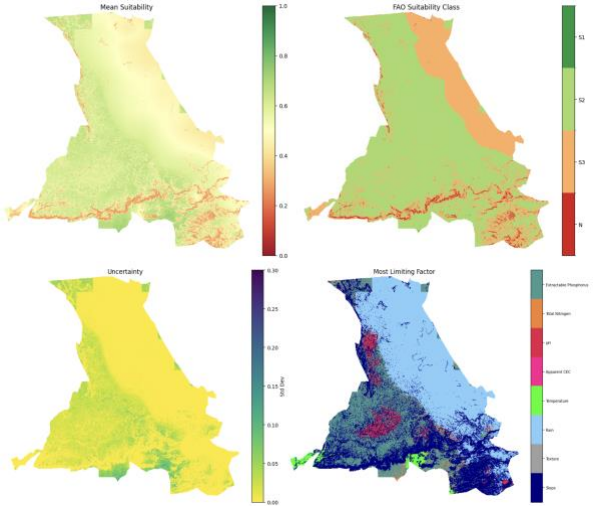


Fig A5. Suitability maps aggregated to ward-level.



Table A5. Fertiliser prescription decision tree. Prescriptions are determined by three ward-level soil parameters: soil pH (threshold 5.5), total nitrogen (TN; threshold 0.2%), and plant-available phosphorus (P; threshold 15 mg kg⁻¹). The eight possible combinations of above/below-threshold values define the primary recommendation. Nutrient rates indicate the equivalent elemental

pH	TN (%)	P (mg kg ⁻¹)	Recommended product(s)	Nutrient rates applied
Acidic soils (pH < 5.5) — all cases require liming in addition to stated product				
< 5.5	< 0.2%	< 15	Option A: MEA Mazao 250 + Manvuno top-dress 185 kg ha ⁻¹ at 6 wks Option B: DAP 100 at planting + CAN 81 kg ha ⁻¹ at 6 wks	N: 39 kg ha ⁻¹ P: 20 kg ha ⁻¹
	< 0.2%	≥ 15	CAN 150 kg ha ⁻¹ (sidedress at 6 wks) + liming	N: 39 kg ha ⁻¹ P: 0
	≥ 0.2%	< 15	TSP 100 kg ha ⁻¹ at planting + liming	N: 0 P: 20 kg ha ⁻¹
	≥ 0.2%	≥ 15	Liming only (no mineral fertiliser)	N: 0 P: 0
Neutral / non-acidic soils (pH ≥ 5.5)				
≥ 5.5	< 0.2%	< 15	DAP 100 kg ha ⁻¹ at planting + Urea 46 kg ha ⁻¹ sidedress at 6 wks	N: 39 kg ha ⁻¹ P: 20 kg ha ⁻¹
	< 0.2%	≥ 15	Urea 85 kg ha ⁻¹ sidedress at 6 wks	N: 39 kg ha ⁻¹ P: 0
	≥ 0.2%	< 15	TSP 100 kg ha ⁻¹ at planting	N: 0 P: 20 kg ha ⁻¹
	≥ 0.2%	≥ 15	No mineral fertiliser required	N: 0 P: 0

pH	TN (%)	P (mg kg ⁻¹)	Recommended product(s)	Nutrient rates applied
Unconditional micro-nutrient checks (applied after main decision tree regardless of pH/TN/P outcome)				
Trigger condition			Recommended action	Rate
Ward mean Zn suitability score < 1.0			Zinc sulfate application; recommend targeted soil sampling	Per soil test
Ward mean K < 0.256 cmol kg ⁻¹			KCl band application at planting	50 kg ha ⁻¹

Source: NAAIAP (2017); product rates are for pure-stand maize.
 Intercropping adjustments: +17 kg ha⁻¹ DAP or TSP for cereal–bean
 intercrop; +27/–22 kg ha⁻¹ Urea for cereal–legume intercrop. Abbreviations:
 DAP = di-ammonium phosphate; CAN = calcium ammonium nitrate;
 TSP = triple superphosphate; KCl = potassium chloride.

[illegible]

Variety		Elevation (m)	Maturity (days)	Prec. (mm)	Yield (t ha ⁻¹)	Key Attributes	Disease Resistance							Stress Tolerance			
Category	Variety						GLS	Blgt	MSV	Rust	E.Rot	FoL	C.Rot	MS	Fl	Ht	Cd
	H515	800–1500	110–130	750–1000	5.3	Tolerant to Leaf Blight, Rust, and GLS; early maturity	✓	✓	✓	—	—	—	—	—	—	—	—
	H511	800–1500	120–135	800–1500	5.0	Standard mid-altitude variety; good for green maize	—	—	—	—	—	—	—	—	—	—	—
Dryland & Lowland	DH01	900–1400	70	200–500	3.3	Early maturity; stay-green trait; tolerant to Rust and Blight	—	✓	—	✓	—	—	—	—	—	—	—
	DH02	900–1400	70–100	200–500	4.0	Moisture-stress tolerant; resistant to MSV; white dent kernels	—	—	✓	—	—	—	—	✓	—	—	—
	DH04	900–1700	75–100	200–500	5.3	Drought tolerant; good standability and husk cover; flood-resistant	—	—	—	—	—	—	—	✓	✓	—	—
	DH08	900–1700	100–120	200–500	5.7	Semi-dent; good standability; tolerant to Ear Rots; flood-resistant	—	—	—	—	✓	—	—	—	✓	—	—
	PH1	0–1200	75–100	400–800	4.4	Heat tolerant; drought resistant; adapted to hot coastal zones	—	—	—	—	—	—	—	✓	—	✓	—
	PH4	0–1200	100–120	400–800	5.7	Very sweet; high prolificacy; Rust and humidity resistant	—	—	—	✓	—	—	—	—	—	—	—
	DH09	1000–1150	105–120	200–500	5.5	Higher yielding than H511; excellent Ear Rot resistance	—	—	—	—	✓	—	—	—	—	—	—
	DH10	800–1400	100–115	200–500	6.6	Out-yields H511 significantly; resistant to lodging and GLS; flood-resistant	✓	—	—	—	—	—	—	—	✓	—	—
	KSDTV-01	0–1000	90–110	250–500	4.2	Fast-maturing hybrid for marginal Eastern and Coast zones	—	—	—	—	—	—	—	—	—	—	—

Fig A6. Example output for the *Long Rains* season in Kapkangani.

The conditions in your area are 68.1% (Moderately Suitable) suitable for maize growth, based on the local soil, weather conditions and terrain.
Elevation: 1748 m | Rainfall: 951 mm | Mean Temperature: 17.8 °C | Max Temperature: 22.8 °C | pH: 5.57 | Nitrogen: 6.195 % | Phosphorus: 18.1 mg/kg

Below are the top 3 recommended seed varieties based on your location.

Variety	Score%	Maturity	Potential Yield (t/ha)	without Fertiliser	with Fertiliser	Key Attributes
H0481	92.5%	160 – 190 Days	11.0	6.6 – 9.35	9.83 – 11.0	High silage potential; lodging and rust resistant; tolerant to GLS
H629	92.5%	160 – 210 Days	10.56	6.34 – 9.98	9.57 – 10.56	Lodging resistant; superior husk cover; resistant to GLS and Ear Rot
H626	54.0%	160 – 210 Days	9.24	5.54 – 7.85	8.77 – 9.24	Stable under cool nights; consistent cob fill even under low light

For the Long Rains season, the optimal planting time for maize in your region is March 25 to April 20.

Below are fertiliser recommendations based on the soil properties in your location. Nitrogen-based fertiliser can improve yields by an average of 2.634 t/ha in your region, while phosphorus-based fertilisers improve yields by 0.640 t/ha.

The soil in your area is low in phosphorus and nitrogen, but the acidity levels are good. For optimal maize growth, it is recommended to apply 100 kg/ha of DAP at planting. Side-dress with the most cost-effective nitrogen source, 46 kg/ha of Urea, six weeks after planting. This will increase both phosphorus and nitrogen concentrations. Please note that if you are intercropping cereal and beans, you should increase DAP by 17 kg/ac; if you are intercropping cereal and legumes, increase DAP by 27 kg/ac and reduce urea by 22 kg/ac.

Your soil may be deficient in zinc and could benefit from zinc sulfate fertiliser. A zinc soil test is recommended.

If a field extension officer is available, we recommend consulting them to refine these guidelines for your specific plot.

⚠ Warning: During the Long Rains season, your area has a historically high flood risk (20.0% of seasons). Avoid basal fertiliser application immediately before forecast heavy rainfall.

The uncertainty of the suitability prediction for this ward is ±1.4%.
